Animal but not dog: Children's computation of implicatures for hierarchically organized categories

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Background

- **Conversational implicatures** (e.g., *some but not all*) arise when a speaker uses a **less informative** expression over **more informative** alternatives[1234].

- **Hierarchically organized common nouns** have a similar informativeness structure:

- If a speaker knows the term ***dog*** but says ***animal***, they might be referring to **another animal → Hierarchical Implicatures (HI).**
- In HI, alternatives are concrete basic-level nouns that are **familiar and contextually accessible**, which might lead to higher success[5].

Research Question

- Can children compute **implicatures with hierarchically organized nouns** at ages where other implicatures are challenging?

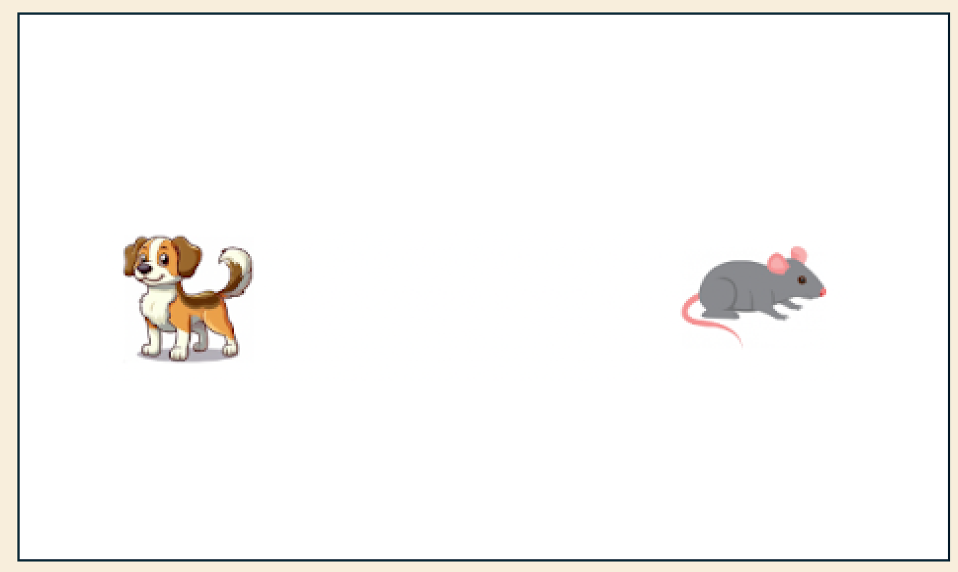
Methods

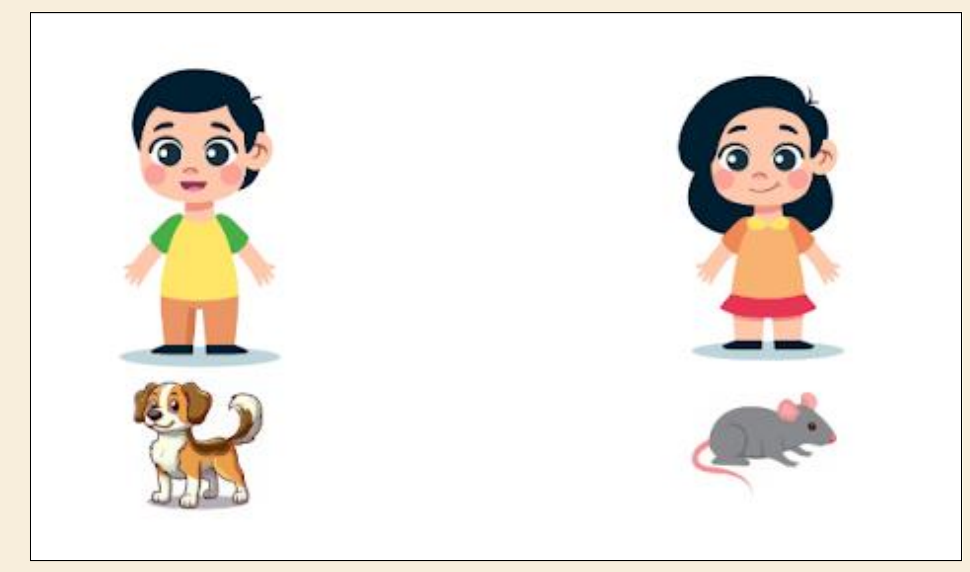
- 120 English-speaking 4;0 – 5;11 yos (M_{age} = 4.82)
- 60 English-speaking adult controls
- **HI task**: 8 2AFC trials
  - Speaker only knows one group of animals
  - **4 control trials:** uses basic-level term.
  - **4 implicature trials**: uses superordinate term *animal.*
- 3 between-subject conditions varying the **number / heterogeneity of animals.**
- **Animal Entailment (AE) task:**

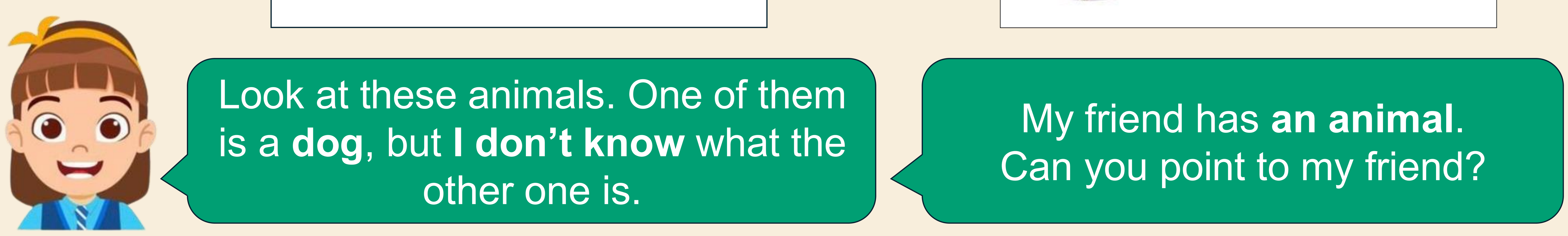
Point to the **animals** / **dogs**.

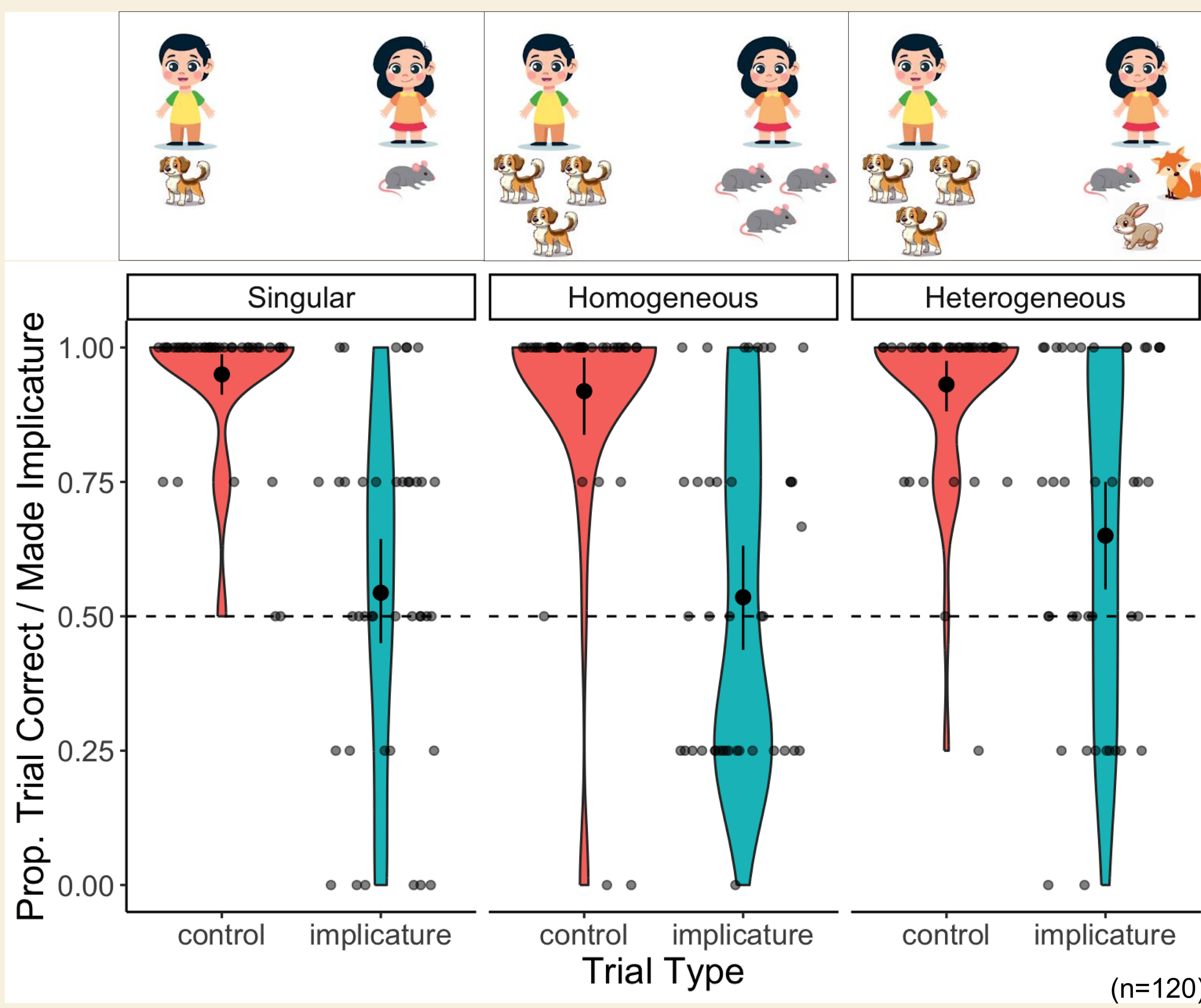
- **Theory of Mind (ToM) task**: 5qs from Wellman & Liu ToM scale[6].

4-5-year-olds compute **conversational implicatures** with **superordinate** and **basic-level** nouns (e.g., *animal but not dog*) in some contexts. Children make these implicatures while reasoning about a speaker's **knowledge of alternative labels**.









Other Results

- **Adults** compute HI (93%) across conditions.
- **Older children** and those with **higher AE / ToM** are more likely to compute HI.

Ongoing & Future Work

- Do children compute HI with **subordinate–basic-level nouns**?
  - Adults (n=30) compute these (71%)
- Does children's **own knowledge** affect HI?

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